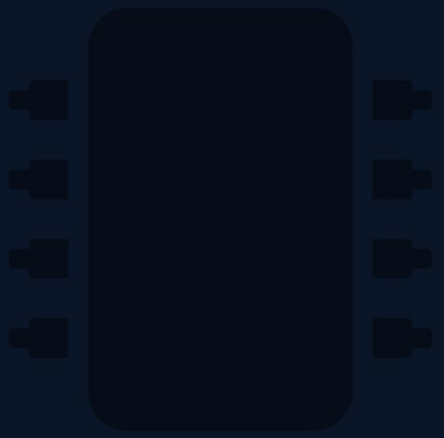
WEEK 2

Software Installation

& First DE0 Test

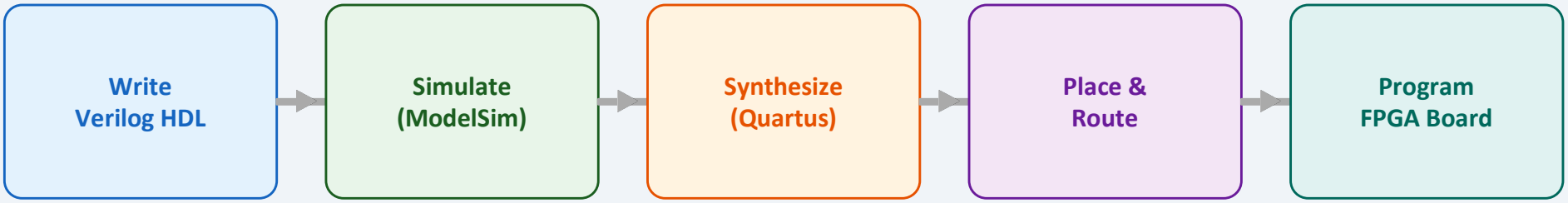
EDA · Quartus II · ModelSim-Altera · USB-Blaster · Hello DE0



INTRO · EDA

Electronic Design Automation (EDA)

Software tools that automate the design, simulation, and verification of electronic circuits



Why do we need EDA?

Without EDA tools, designing a chip with millions of gates by hand would take decades. EDA automates repetitive tasks — wiring, timing checks, layout — so engineers can focus on architecture and innovation.

EDA Tools in this course

Quartus II — the main EDA tool for synthesis, fitting, and FPGA programming.
ModelSim-Altera — functional simulation before touching hardware.

INTRO · FPGA

What is an FPGA?

Field-Programmable Gate Array — a chip whose logic can be reconfigured after manufacturing



Reconfigurable Hardware

Unlike an ASIC, an FPGA can be reprogrammed thousands of times. Perfect for prototyping and research.



Parallel Execution

Hardware runs all logic simultaneously. An FPGA can process many operations in a single clock cycle.



Low Latency

No OS overhead. Deterministic nanosecond-level response — ideal for real-time control and DSP.



LUTs, Flip-Flops & DSP Blocks

Built from Look-Up Tables (LUTs), registers, memory blocks, and hard DSP multipliers.

INTRO · FPGA vs CPU vs ASIC

FPGA vs CPU vs ASIC

	CPU	FPGA	ASIC
Architecture	Sequential, fixed cores	Reconfigurable logic fabric	Fixed custom silicon
Performance	Moderate	High — parallel hardware	Highest
Flexibility	Software only	Hardware reprogrammable	None after fab
Development time	Fast	Days–weeks	Months–years
Unit cost	Low (mass prod.)	Medium	Very low (volume)
Power	Moderate	Lower than CPU for same task	Lowest
Best for	General computing	Prototyping, real-time DSP	Mass production chips

INTRO · FPGA BRANDS

Commercial FPGA Brands

Three major vendors dominate the market — each with distinct strengths

Used in this course

Intel / Altera

Product Families:

Cyclone, Arria, Stratix, Agilex

EDA Tools:

Quartus Prime / Quartus II

★ *Cyclone III EP3C16F484C6 — the chip on our DE0 board*

Industry leader

AMD / Xilinx

Product Families:

Spartan, Artix, Kintex, Virtex, Versal

EDA Tools:

Vivado, ISE, Vitis

★ *Most widely used in industry & academia worldwide*

Niche / Low power

Microchip / Microsemi / Lattice

Product Families:

PolarFire, iCE40, ECP5, IGLOO2

EDA Tools:

Libero SoC, Radiant, Diamond

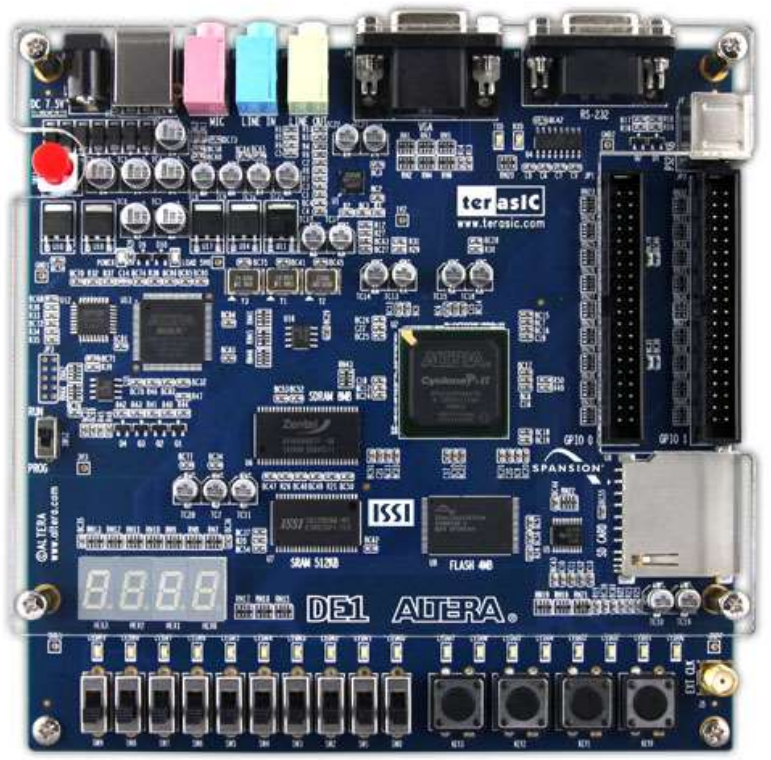
★ *Low power, security-focused; popular in IoT and space*

INTRO · DE0 BOARD

Altera DE1 Development Board

Board Specifications

FPGA	Cyclone III EP3C16F484C6
Logic Elements	15,408 LEs
Memory	504 Kbits embedded RAM
PLLs	4 PLLs
Clock	50 MHz oscillator (CLOCK_50)
Switches	10 × Slide Switch (SW[9:0])
LEDs	10 × Green LED (LEDG[9:0])
7-Seg Display	4 × 7-segment (HEX0–HEX3)
Keys	3 × Push Button (KEY[2:0])
GPIO	2 × 40-pin expansion header



ARCHITECTURE · OVERVIEW

Cyclone Architecture — Overall Block Diagram

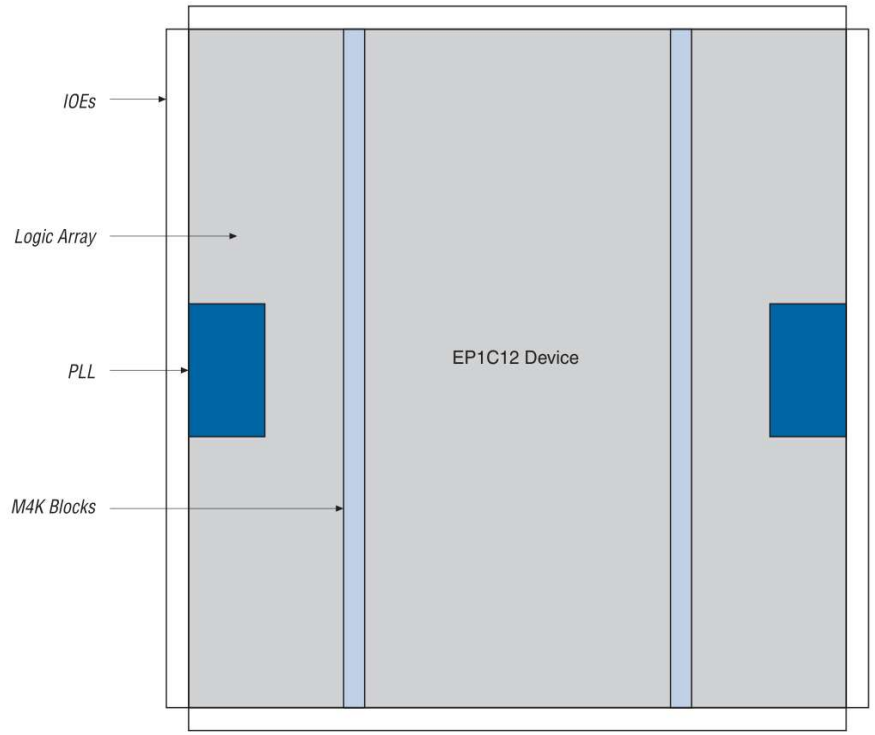


Figure 2–1: Cyclone EP1C12 Block Diagram

Logic Array (LABs)

2D row/column array of Logic Array Blocks (LABs). Each LAB contains 10 Logic Elements (LEs). Range: 2,910–20,060 LEs.

M4K Embedded RAM

True dual-port RAM blocks (4,608 bits each). Run up to 250 MHz. Used for ROM, FIFO, SRAM in your Verilog design.

Phase-Locked Loops (PLLs)

Up to 2 PLLs per device. Clock multiplication, division, and phase shifting. Essential for multi-clock designs.

I/O Elements (IOEs)

Located on device periphery. Each IOE has input, output, and OE registers. Supports LVTTL, LVCMOS, LVDS, PCI, SSTL standards.

LAB · LOGIC ARRAY BLOCK

Logic Array Block (LAB)

The fundamental grouping unit — each LAB contains 10 Logic Elements plus shared interconnect resources

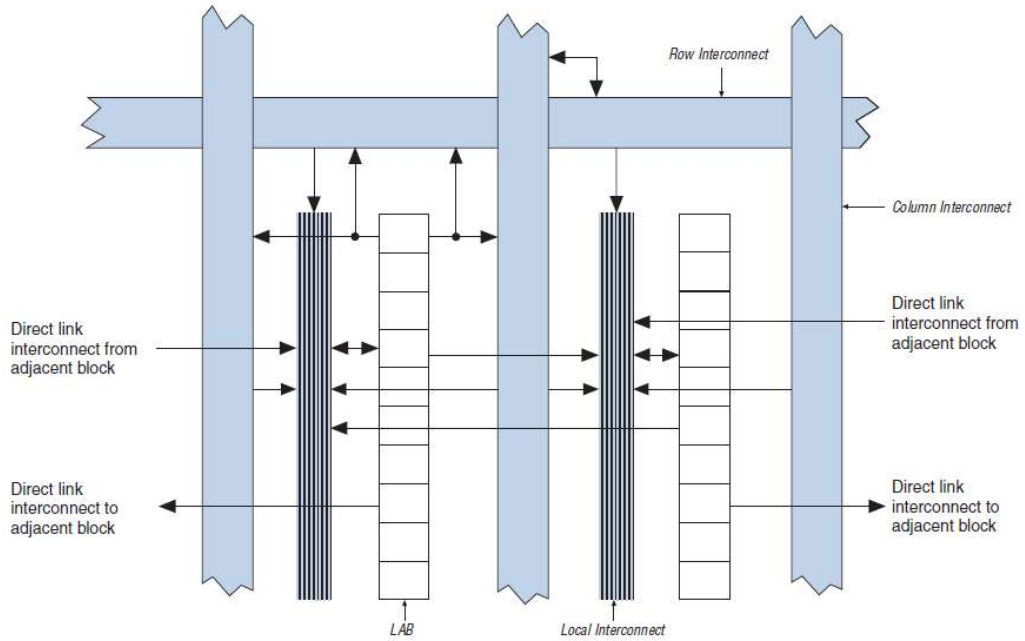


Figure 2–2: Cyclone LAB Structure (Source: Altera)

10 LEs per LAB

Each LAB holds exactly 10 LEs sharing local interconnect, clock, and control signals.

Local Interconnect

Transfers signals between LEs inside the same LAB — faster than using row/column routing.

LUT Chain

Passes the combinatorial output of one LE directly to the adjacent LE for wide logic functions.

Register Chain

Connects registers of adjacent LEs inside a LAB — ideal for fast shift-register implementations.

Direct Link

Bypasses row/column for fast LAB-to-LAB connections between neighboring blocks (left/right).

LE · LOGIC ELEMENT

Logic Element (LE) — The Smallest Unit of Logic

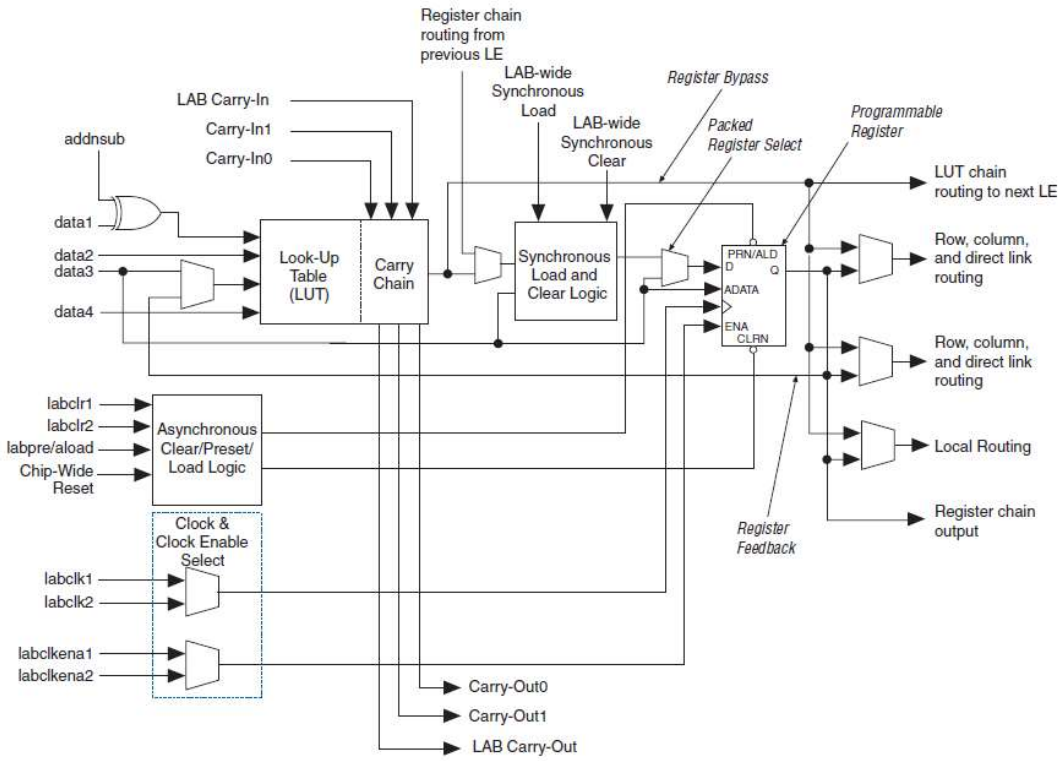


Figure 2–5: Cyclone LE Internal Structure (Source: Altera)

4-Input LUT

Implements ANY logic function of 4 variables. This is where your Verilog combinational logic lives.

Carry Chain

Fast adder carry-select chain for arithmetic. Enables high-speed adders, counters, comparators.

Programmable Register

D/T/JK/SR flip-flop. Clocked by LAB-wide clock. Stores state in sequential logic (always @posedge).

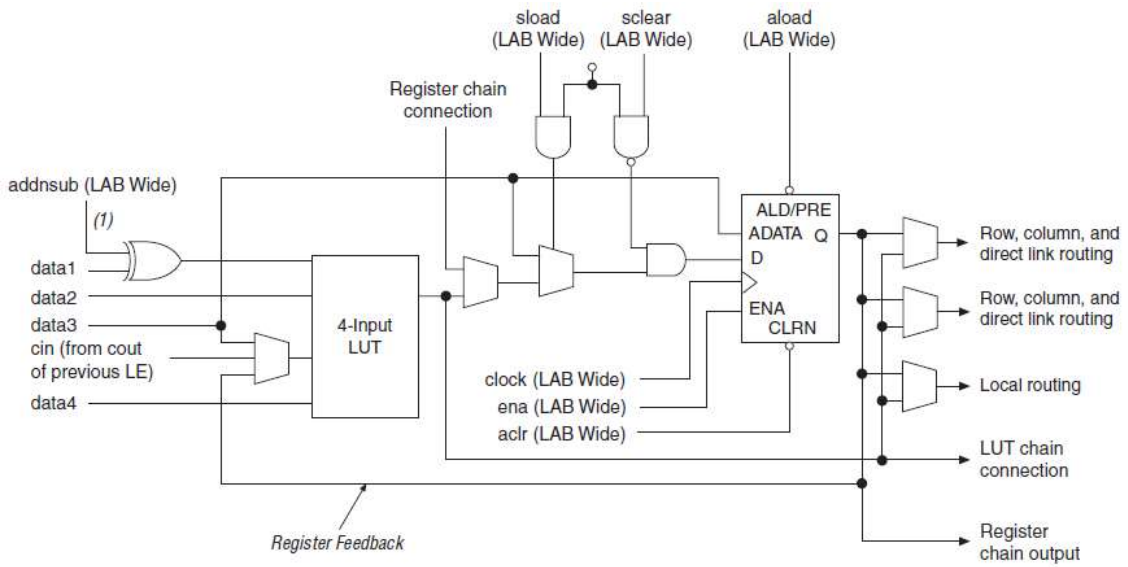
3 Outputs

LE drives local, row/column, and direct-link interconnects — LUT and register can drive independently (register packing).

LE MODE 1 · NORMAL MODE

LE Normal Mode — General Combinational Logic

Used for: gates, muxes, decoders, state machines — any function of ≤ 4 variables



4-Input LUT

Implements any Boolean function of data1–data4. In normal mode, Quartus may also select carry-in as one of the four inputs for adder tail-end.

Register Bypass (combinational path)

For pure combinational logic (assign statements), the LUT output bypasses the D flip-flop and drives the LE outputs directly — no clock needed.

Packed Register

LUT and register can implement two unrelated functions simultaneously. This doubles resource efficiency — Quartus does this automatically.

LUT Chain → next LE

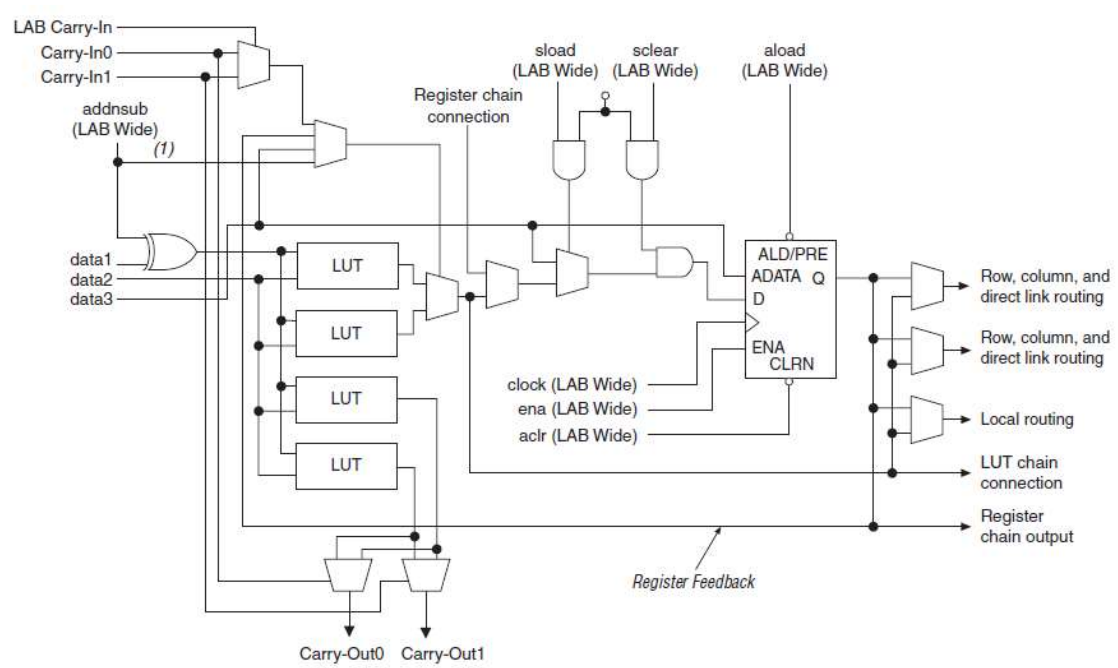
The combinational output can cascade directly to the adjacent LE's input for fast wide fan-in logic (up to 10 LEs in one LAB).

```
assign Y = (A & B) | (~C & D);  // → one LE, normal mode
```

LE MODE 2 · DYNAMIC ARITHMETIC MODE

LE Dynamic Arithmetic Mode — Fast Adders & Counters

Used for: adders, subtractors, counters, accumulators, comparators, wide parity functions



Four 2-Input LUTs

Instead of one 4-input LUT, the LE uses four 2-input LUTs. Two compute possible sums (carry-in=0 and carry-in=1 in parallel); two generate carry-outs.

Carry-Select Chain

Pre-computes both carry-in=0 and carry-in=1 outcomes simultaneously. LAB carry-in selects the correct result — only selection delay is in the critical path.

addnsub (LAB-wide)

A single LAB-wide control signal switches all LEs between A+B and A−B. Subtraction is done by two's complement (invert B + set carry-in=1). No extra LEs needed.

Clock Enable & Sync Load

Supports synchronous up/down count, synchronous clear, and synchronous load — all LAB-wide. Quartus uses this mode automatically for counters and LPM functions.

```
always @(posedge clk) cnt <= cnt + 1;  // → arithmetic mode
```

MEMORY · M4K RAM BLOCKS

M4K Embedded RAM Blocks

True dual-port synchronous memory blocks embedded in columns between the LAB array

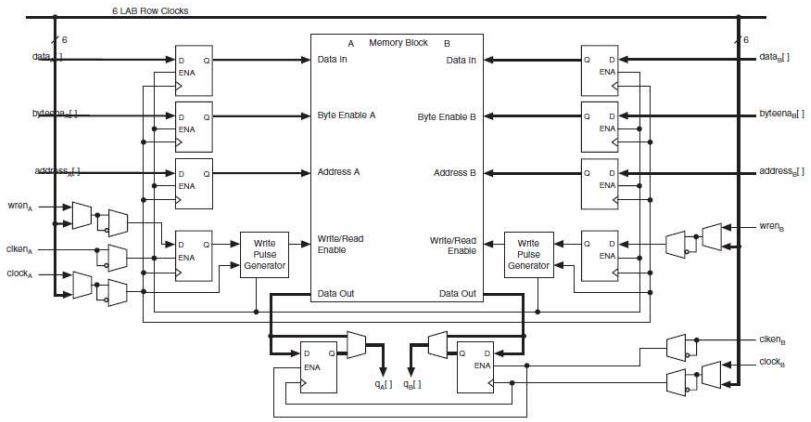


Figure 2–12: True Dual-Port Memory (Source: Altera)

M4K Specifications

4,608 bits per block · Up to 250 MHz · EP1C12: 52 blocks (2 columns) · Configs: 4K×1 → 128×36

True Dual-Port

Two independent ports (A+B). Each port can read or write simultaneously at different clock frequencies. Most flexible mode.

Simple Dual-Port

One write port, one read port. Simultaneous read and write. Classic FIFO/buffer pattern.

Single-Port

Non-simultaneous read/write via shared port. Two single-port blocks can share one M4K if each ≤ 2K bits.

Shift Register Mode

Up to 4,608-bit shift register. Saves LEs vs flip-flop chains. Used in DSP filters, correlators.

ROM Mode

Pre-loaded at synthesis using .mif initialization file. Read-only after configuration.

FIFO Buffer

Built-in FIFO controller. Separate read/write clocks (DCFIFO). Full/empty flags included.

Global Clock Network & Phase-Locked Loops (PLLs)

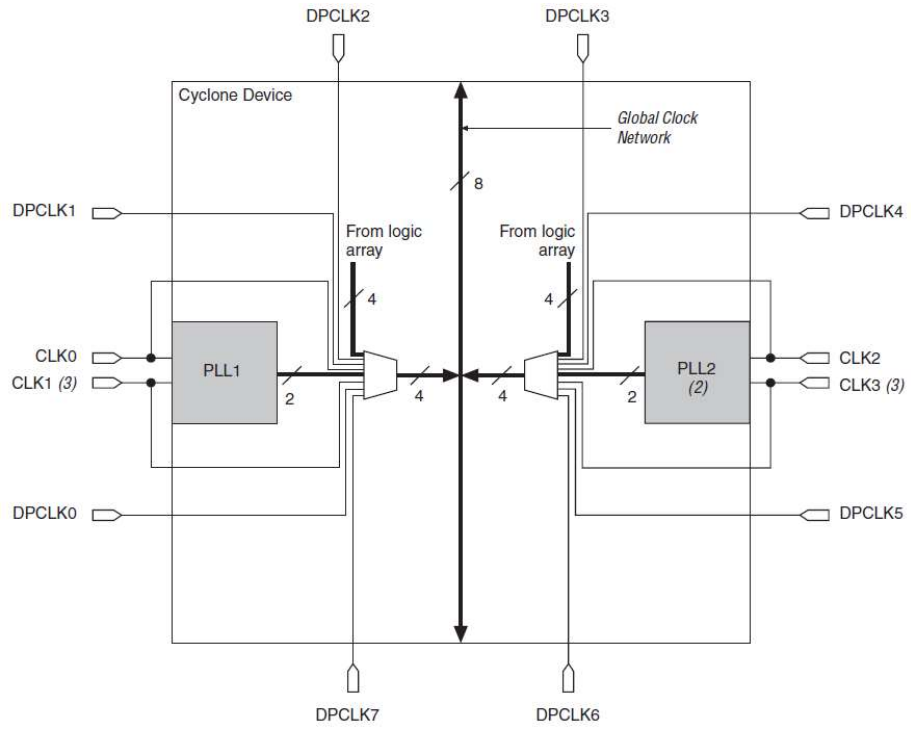


Figure 2–22: Global Clock Generation (Source: Altera)

8 Global Clock Lines

8 dedicated low-skew lines drive the entire chip — IOEs, LEs, and M4K blocks. Also used for clock enables and async clears.

2 PLLs (PLL1 & PLL2)

Multiply, divide, and phase-shift input clocks. m/n × post-scale counter formula. Minimum phase step: 125 ps.

4 Dedicated CLK Pins

CLK[3:0] drive the PLLs directly. Also 8 dual-purpose DPCLK pins for high-fanout control signals.

In This Course

CLOCK_50 (50 MHz) feeds the global clock network → clocks all your flip-flops via always @(posedge CLOCK_50).

I/O · IOE STRUCTURE

I/O Element (IOE) — Peripheral Interface

Each I/O pin on the DE0 board is controlled by one IOE — a bidirectional buffer with 3 registers

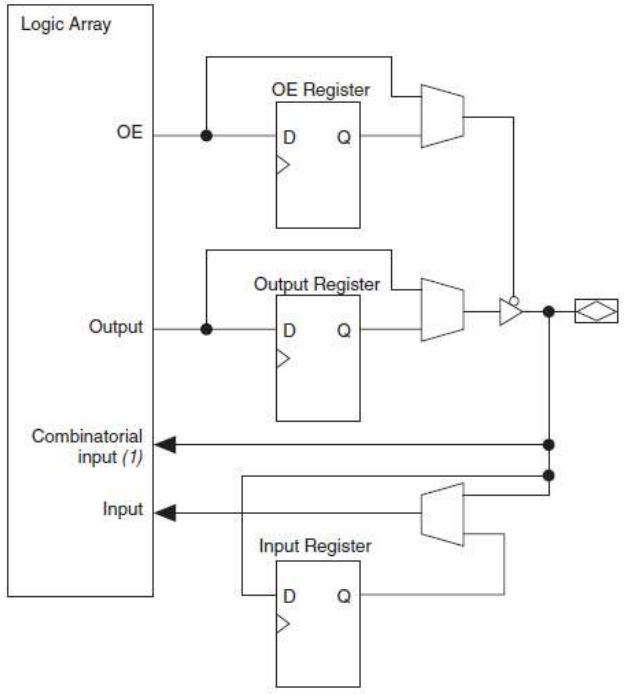


Figure 2–27: Cyclone IOE Structure (Source: Altera)

Input Register
Registers incoming data from the pin for fast setup-time performance. Has its own clock, separate from the output.

Output Register
Registers outgoing data for fast clock-to-output timing. Shares clock source with OE register.

OE (Output Enable) Register
Controls the tri-state buffer direction. Registered for fast clock-to-output-enable timing — enables half-duplex.

Programmable Features
Drive strength (4–24 mA), slew rate control, bus-hold, pull-up resistor, open-drain output, programmable delays.

I/O Standards Supported
LVTTL/LVCMOS (1.5–3.3 V), PCI 3.3 V, LVDS (640 Mbps), SSTL-2/3. Each I/O bank has its own VCCIO supply.

SUMMARY · VERILOG ↔ HARDWARE

Verilog → Cyclone Hardware Mapping

How your Verilog code maps onto physical Cyclone resources after synthesis

Verilog Construct	Maps to Hardware	Physical Location
`assign Y = A & B & C;`	**LUT (Normal Mode)**	Inside an LE
`always @(posedge clk)`	**Flip-Flop Register**	LE register
`4-bit counter / adder`	**Carry Chain + Arithmetic Mode**	Multiple LEs in LAB
`reg [7:0] mem [255:0]`	**M4K RAM Block**	M4K column
`input/output ports`	**I/O Element (IOE)**	Device periphery
`CLOCK_50`	**Global Clock Network**	8 global clock lines

Quartus II automatically places, routes, and optimizes — your job is to write clean, synthesizable Verilog!

TOOL · QUARTUS II

Altera Quartus II — EDA Tool Overview

The complete design environment for Intel/Altera FPGAs: from HDL to bitstream



Version used in this course

Quartus II 13.0sp1 Web Edition
Free — no license required
Supports Cyclone III (DE0 chip)

ModelSim-Altera integration

Built-in Simulation → ModelSim-Altera
NativLink auto-compiles testbench
Waveform viewer included

Programmer tool

Tools → Programmer
USB-Blaster interface
Supports .sof (SRAM) & .pof (Flash)

Today's Lab Agenda

01



Install Quartus II

Download and install Quartus II 13.0sp1 Web Edition — the main EDA tool for this course

02



Install ModelSim-Altera

Install the HDL simulator — used to verify logic before programming the FPGA board

03



Install USB-Blaster Driver

Configure the JTAG interface driver so Quartus can communicate with the DE0 board

04



Hello DE0 — First Project

Create a Quartus project: assign SW[9:0] → LEDG[9:0], compile, program, and test on DE0

01 · QUARTUS II

Install Quartus II Web Edition

EDA Tool: Design → Synthesis → Place & Route → Program

1

Download Quartus II 13.0sp1 Web Edition
altera.com/products/software/quartus-ii/subscription-edition.html

2

Choose installer for your OS:
Windows → QuartusSetupWeb-13.0.1.232.exe
Linux → QuartusSetupWeb-13.0.1.232.run

3

Run installer → check ✓ Cyclone III device support
Click Next on every page → use default installation path

4

After install → open Quartus II → Help → License Setup
Select Free Web Edition License (no registration needed)

02 · MODELSIM-ALTERA

Install ModelSim-Altera Starter Edition

1

Download ModelSim-Altera 10.1d Starter Edition
from altera.com (or use the Download Center within Quartus II)

2

Run installer → install in the same root folder as Quartus II
e.g., C:\altera\13.0sp1\modelsim_ase

3

Open ModelSim → test: File → New → Source → Verilog
Type: module test; initial $display("OK"); endmodule

4

Linking with Quartus II:
Tools → Options → EDA Tool Options → set ModelSim-Altera path

ModelSim Starter Edition has a simulation speed limit — but it is more than sufficient for every exercise in this course

03 · USB-BLASTER DRIVER

Install USB-Blaster Driver

Connect PC ↔ DE0 Board via JTAG interface

Windows

1. Connect a USB cable from PC → USB-Blaster port on DE0
2. Windows will show "Found New Hardware"
3. Device Manager → Other Devices → USB-Blaster → Right-click → Update Driver
4. Browse → C:\altera\13.0sp1\quartus\drivers\usb-blaster → Install driver
5. Verify: "Altera USB-Blaster" appears in Device Manager

Linux / macOS

Linux: driver is bundled with Quartus II — no separate install.
Run: sudo chmod a+rw /dev/bus/usb/*/*
Or add yourself to the plugdev group.

✅ Verify the driver works

1. Open Quartus II
2. Tools → Programmer
3. Hardware Setup → "USB-Blaster" must appear in the list
4. If not shown → restart PC and try again

04 · HELLO DE0

Create Your First Quartus Project — Hello DE0

1 New Project Wizard

File → New Project Wizard
Project name: hello_de0
Top-level entity: hello_de0

2 Select Device

Family: Cyclone II
Device: EP2C20f4847n
(matches the DE1 board chip)

3 Create Verilog File

File → New → Verilog HDL File
Save as hello_de0.v
Write the module code

4 Pin Assignment

Assignments → Pin Planner
Assign SW[0] → PIN_J6
Assign LEDG[0] → PIN_J1

5 Compile

Processing → Start Compilation
Wait 2-5 minutes
Check: 0 errors

6 Program DE0

Tools → Programmer
Select .sof file
Click Start → Programming Done!

04 · VERILOG CODE

Verilog Code — Hello DE0

VERILOG

```
module hello_de0 (
  input
  [9:0] SW,
  output
 [9:0] LEDG
);


  assign
 LEDG = SW;  // SW mirrors LEDG

endmodule
```

module / endmodule

Defines the boundary of a circuit — like a function in C but describes hardware structure, not a program.

input [9:0] SW

A 10-bit input port connected to the 10 slide switches on the DE0 board.

output [9:0] LEDG

A 10-bit output port driving the 10 green LEDs on the DE0 board.

assign LEDG = SW

Continuous assignment — a direct wire connection. When a switch is flipped, the LED responds immediately.

04 · PIN ASSIGNMENT

Pin Assignment — DE0 Board

Map Verilog ports to physical pins of Cyclone III EP3C16F484C6

Signal	Direction	FPGA Pin	DE0 Component
SW[0]	Input	PIN_J6	Slide Switch 0
SW[1]	Input	PIN_K6	Slide Switch 1
SW[2]	Input	PIN_K5	Slide Switch 2
SW[3]	Input	PIN_H5	Slide Switch 3
LEDG[0]	Output	PIN_J1	Green LED 0
LEDG[1]	Output	PIN_J2	Green LED 1
LEDG[2]	Output	PIN_J3	Green LED 2
LEDG[3]	Output	PIN_J4	Green LED 3
SW[9:4]…	Input	…	Slide Switch 4–9 (see DE0 manual)
LEDG[9:4]…	Output	…	Green LED 4–9 (see DE0 manual)

AI ASSISTANT · GEMINI

Using Gemini as Your AI Assistant in Week 2

💬 Generate your first Verilog

"Write a simple Verilog module for DE0: SW[0] → LEDG[0] with inline comments"

🔧 Debug USB-Blaster issues

"Quartus Programmer cannot detect USB-Blaster running on Windows 11 — what should I do?"

📖 Understand key concepts

"What is the difference between wire and reg in Verilog? Explain with a simple example."

❓ Fix error messages

*"Error: Port SW not found in module
What is wrong with this code? [paste code]"*

Week 2 — Summary

- Installed Quartus II 13.0sp1 Web Edition ✓
- Installed ModelSim-Altera Starter Edition ✓
- Installed USB-Blaster JTAG Driver ✓
- Created first Quartus project & assigned FPGA pins ✓
- Compiled & programmed DE0 — SW[9:0] → LEDG[9:0] ✓
- Used Gemini AI to generate code, explain concepts & debug ✓

▶ Week 3 : Verilog Fundamentals I — Module, Port, Data Types · Half Adder & Full Adder on DE0